

LEFT-OUT SECTION FOLLOWS.

♀/ 4K SYSTEM GENERATION METHOD.

/ THIS METHOD HAS NOT BEEN IMPLEMENTED TO DATE. IT IS THEORETICALLY POSSIBLE TO
/ CREATE AN ALTERNATE SYSTEM GENERATION METHOD TO WRITE OUT SMALLER SECTIONS OF
/ THE OPERATING SYSTEM, EACH MODULE WITH ITS OWN GENERATION PROGRAM. [NOTE:
/ CERTAIN SYSTEM DEVICE HANDLERS WILL REQUIRE AT LEAST 8K TO FUNCTION; IT IS NOT
/ CLEAR IF THESE CASES CAN BE HANDLED ON A 4K SYSTEM.] IT IS POSSIBLE TO DESIGN
/ A GENERIC SYSTEM BUILDER PROGRAM FOR THE PURPOSE OF DEALING WITH THE CUSTOM
/ LOGISTICS ASSOCIATED WITH EACH SYSTEM COMPONENT. [NOTE: SYSTEM PROGRAMS WITH
/ DEFINED OVERLAY STRUCTURES WILL PRESENT AN ADDITIONAL CHALLENGE TO SUCH A
/ PROGRAM.]

/ FUTURE SYSTEM GENERATION ISSUES.

/ ANY FUTURE SYSTEM GENERATION UTILITY SHOULD EMBRACE BOTH METHODS OF CREATING A
/ STARTER SYSTEM AND ADDITIONAL COMPONENTS. BY PROPER DESIGN OF THE BINARY
/ MODULES, THE DIRECT LOADING APPROACH USED IN THE HISTORIC METHOD CAN BE
/ CONTINUED WITH LITTLE CHANGE TO THE SOURCE CODE AS WELL AS USING A CUSTOM
/ GENERATION PROGRAM DESIGNED TO RUN ON 4K MACHINES TO THE EXTENT POSSIBLE. THE
/ GENERATION UTILITY CAN BE LOADED INDEPENDENTLY INTO A 4K SYSTEM AND THEN DRAW
/ UPON SEVERAL POSSIBLE RESOURCES TO WRITE OUT PORTIONS OF THE OPERATING SYSTEM.

/ THIS WOULD INCLUDE DIRECTLY HANDLING BINARY PAPER TAPES IN A MANNER AKIN TO
/ THE STANDARD BINARY PAPER TAPE LOADER WITH SOME EXTENSIONS: DESIGNATED SETS OF
/ SPECIAL ORIGIN SETTINGS CAN GUIDE THE PROCESS IN VARIOUS WAYS. THIS WOULD
/ ALLOW DIRECT BINARY LOADING METHODS TO FUNCTION AS IN THE HISTORIC METHOD.
/ THE UTILITY WOULD RECOGNIZE SPECIAL LOADING CODES TO REDIRECT FIELD SETTINGS
/ TO COMMON LOADING IN FIELD 0 BY THE GENERATION PROGRAM TO ALLOW OVERLAY
/ STRUCTURES TO BE WRITTEN OUT WITHOUT THE USE OF EXTENDED MEMORY.

/ OTHER METHODS COULD INCLUDE EXTENDED MEMORY LOADING OF CERTAIN COMPONENTS
/ ALONG WITH THE GENERATION PROGRAM IN FIELD 0. THIS WOULD ALLOW LESS
/ CUMBERSOME SYSTEM CREATION WITH NO DEPENDENCY ON PAPER TAPES [OR EVEN AN
/ OPERATING SYSTEM]. RUNNING UNDER AN EXISTENT P?S/8 SYSTEM MIGHT ALSO ADD SOME
/ ADDITIONAL LOADING METHODS. REGARDLESS OF METHOD CHOSEN, ALL CASES MUST BE
/ HANDLED TO BE EFFECTIVE.

/ ADDITIONAL ISSUES.

/ WRITING OUT BINARY FILES TO DESIGNATED BLOCKS IS THE PRIMARY PURPOSE OF A
/ GENERATION PROGRAM. HOWEVER, VARIOUS P?S/8 STRUCTURES MUST BE CREATED AS WELL
/ SUCH AS THE TINY FILE SYSTEM [TFS] DIRECTORY [UNLESS IT PRE-EXISTS IN WHICH
/ CASE IT COULD BE VALIDATED TO PERMIT RETAINING ANY EXISTENT USER FILES]. A
/ PROPER UTILITY SHOULD BE ABLE TO ADJUST ALL DIRECTORY STRUCTURES AS NECESSARY.

/ FOR SOME STRUCTURES, USER INPUT MIGHT BE REQUIRED TO CUSTOMIZE DIRECTORY
/ PARTICULARS AND RELATED MATTERS. SPECIFIC SYSTEM DEVICE HANDLERS CAN SUPPLY
/ PARTIAL INFORMATION SUCH AS MAXIMUM ADDRESSABLE BLOCK ON THE DEVICE, ETC. TO
/ AID IN THE DIRECTORY SETUP PROCESS. NOTE: P?S/8 DOES NOT USE A CENTRAL
/ ALL-PURPOSE DIRECTORY STRUCTURE [OUTSIDE OF THE SHELL ENVIRONMENT]. AS SUCH,
/ EACH INDIVIDUAL DIRECTORY MUST BE CUSTOMIZED.]

END OF LEFT-OUT SECTION PART 1.

♀/ SOURCE CODE MODIFICATIONS TO ACCOMMODATE A SYSTEM GENERATION PROGRAM.

/ ACCOMMODATING THE DUAL GENERATION APPROACH SUGGESTED ABOVE WILL REQUIRE
/ INSERTION OF SPECIAL LOADING INFORMATION INTO THE CODE WITHOUT CHANGING THE
/ OVERALL BINARY OUTPUT AS DICTATED BY THE HISTORIC LOADING METHOD. TO
/ ACCOMPLISH THIS, HEADER INFORMATION [ANALOGOUS TO THE SIMPLISTIC DTORG
/ FUNCTION ASSOCIATED WITH TYPESET-8] CAN BE INSERTED BY MERELY DEFINING SPECIAL
/ INFORMATION IMPLEMENTED AS A SERIES OF ORIGIN SETTINGS NOT DIRECTLY ASSOCIATED
/ WITH ANY ACTUAL LOADABLE BINARY DATA WORDS.

/ FOR EXAMPLE, A SPECIAL ORIGIN GROUPING COULD INDICATE AN ABSOLUTE SYSTEM BLOCK
/ THE CODE FOLLOWING THE GROUP SHOULD BE WRITTEN TO. OTHER COMBINATIONS COULD
/ INDICATE PROGRAM-GENERATED PARAMETERS SUCH AS THE LARGEST EXISTENT BLOCK ON A
/ SYSTEM DEVICE TO BE USED IN CONJUNCTION WITH OTHER INFORMATION TO SETUP THE
/ SYSTEM DIRECTORY STRUCTURES. ANY NUMBER OF ADDITIONAL PARAMETERS COULD BE
/ DEFINED FOR THE BENEFIT OF THE SYSTEM GENERATION PROGRAM WITHOUT AFFECTING THE
/ DIRECT BINARY LOADING OF THE HISTORIC SYSTEM GENERATION METHOD.

/ AS MENTIONED EARLIER, THERE IS A NEED TO DEFINE INTERCEPT ROUTINES TO ALLOW
/ THE LOGICAL CONSOLE OVERLAY TO PREVENT HALT ON ERRORS IN THE SYSTEM DEVICE
/ HANDLER. CERTAIN SYSTEM DEVICE HANDLERS ALSO DEFINE CUSTOM INTERCEPTS FOR THE
/ EXPRESS PURPOSE OF IMPROVING PERFORMANCE.

/ IN THE CASE OF THE TC01/TC08 SYSTEM DEVICE HANDLER, IT IS POSSIBLE TO DEFINE
/ A MODIFICATION TO THE SEARCH CODING TO SUPPORT INDIVIDUALIZED LAST TRANSFERRED
/ BLOCK VALUES FOR EACH OF THE 8 DRIVE UNITS 0 THROUGH 7. THIS WILL ENHANCE
/ SYSTEM PERFORMANCE WHEN ACCESSING MULTIPLE DRIVES. [NOTE: WITHOUT THE
/ CODE MODIFICATION, SELECTING MULTIPLE DRIVE UNITS MAY CAUSE THE INITIAL SEARCH
/ TO BE PERFORMED IN THE NON-OPTIMAL DIRECTION; THIS IS INSTANTLY CORRECTED
/ AFTER THE FIRST BLOCK MARK IS DETECTED. HOWEVER, WITH THE ENHANCEMENTS IN
/ PLACE THIS SLIGHT POTENTIAL DEFICIENCY IS ELIMINATED.]

/ AS CURRENTLY IMPLEMENTED, A BUFFER IS RESERVED FOR THE VARIOUS SYSTEM HANDLER
/ INTERCEPT ROUTINES. THIS INCLUDES APPROPRIATE HANDLER PATCHING CODE AND CODE
/ TO BE ADDED TO THE LOGICAL CONSOLE OVERLAY. UNIQUE ERROR MESSAGE STRINGS ARE
/ PROVIDED TO CREATE OPTIONAL VERBOSE ERROR MESSAGES DURING ERROR RECOVERY.
/ USING SPECIAL-PURPOSE ORIGIN GROUPS, THESE ADDITIONAL COMPONENTS CAN BE
/ LOADED BY THE SYSTEM GENERATION PROGRAM WITHOUT REQUIRING EXTENDED MEMORY.

/ FOR MANY DEVICES, THE COMPONENTS OF THE LEGENDARY BUFFERLESS BINARY LOADER
/ [KNOWN AS "SLURP" AS COINED BY ITS INVENTOR, THE EQUALLY LEGENDARY RICHARD
/ LARY] AND ITS CUSTOM SYSTEM DEVICE HANDLER RELOADER [SLURP-COMPATIBLE INTERNAL
/ COMPONENT OF THE P?S/8 "BIN" PROGRAM INVOKED BY USING THE /I OPTION SWITCH]
/ CAN ALSO BE LOADED BY THIS METHOD WITHOUT INTERFERING WITH THE HISTORIC
/ GENERATION METHOD. [NOTE: THE SLURP LOADER AND SYSTEM DEVICE HANDLER RELOADER
/ SPECIFIC TO THE TC01/TC08 ARE NOT INCLUDED HERE. FUTURE RELEASES MAY ADD
/ THESE COMPONENTS ONCE FORMALIZATION OF CERTAIN SYSTEM GENERATION ISSUES HAS
/ BEEN COMPLETED.]

/ ?/ WRITING OUT THE SYSTEM HANDLER AND OTHER COMPONENTS OF P?S/8 REQUIRES A
/ COLLECTION OF BUFFERS IN THE USER SPACE. THIS ALLOWS ANY CONFORMING OPERATING
/ SYSTEM TO LOAD THE STARTER BINARY OPERATING SYSTEM. AS SUCH, ALL SUCH BUFFERS
/ MUST BE DEFINED IN THE RANGE OF 00000-07577 AS ALL RELEVANT OPERATING SYSTEMS
/ RESERVE THE LAST PAGE OF FIELD 0, INCLUDING THE PAPER-TAPE BINARY LOADER.

/ PART OF THE LOADED CODE MUST BE A ROUTINE TO WRITE OUT ALL COMPONENTS FROM
/ MEMORY TO THE VARIOUS BLOCKS ASSIGNED; LATER THE SYSTEM BOOTSTRAP PROCESS WILL
/ BRING INTO MEMORY WHATEVER PORTIONS ARE NECESSARY AT ANY GIVEN TIME.

/ IN THIS PARTICULAR SYSTEM DEVICE HANDLER DESCRIPTION, AN UNSTATED MEANS MUST
/ BE PROVIDED TO ACCOMPLISH THE ABOVE-MENTIONED WRITE OPERATIONS. [THIS IS
/ HANDLED ELSEWHERE IN THE P?S/8 SYSTEM GENERATION PROCESS.] VARIOUS DESIGN
/ DECISIONS MUST BE MADE. HISTORICALLY THIS HAS REQUIRED A 16K MACHINE TO WRITE
/ OUT ALL RELEVANT BINARY DATA THAT COULD FIT INTO X0000-X7577, WHERE X IS 0, 1,
/ 2 AND 3. [THIS SPECIFIC COLLECTION OF BINARY FILES COULD BE LOADED BY PAPER
/ TAPE, P?S/8 OR OS/8.] PART OF THE BINARY LOAD IS A ROUTINE WHICH CAN WRITE TO
/ THE SYSTEM DEVICE, AND A LIST OF SPECIFIC OPERATIONS TO PERFORM.

/ IN THE CASE OF THE TC01/TC08, THE ORIGINAL METHOD USED WAS TO MOVE THE IMAGE
/ OF THE SYSTEM DEVICE HANDLER AS DEFINED HERE TO ITS DESIGNATED WORKING ADDRESS
/ RANGE OF 07600-07777; THIS WILL FUNCTION PROPERLY BECAUSE THE CODE CREATED TO
/ EXECUTE THERE IS NORMALLY LOADED THERE UPON SYSTEM BOOTUP.

/ UNFORTUNATELY, THIS IS NOT ALWAYS THE CASE. SEVERAL OTHER SYSTEM DEVICE
/ HANDLERS ARE NOT CAPABLE OF FUNCTIONING MERELY BY CODE RELOCATION ALONE.

/ FOR CERTAIN DEVICES, BOOTSTRAP CONVENTIONS MAY REQUIRE THE CONTENTS OF LOGICAL
/ BLOCK 0000 TO BE SIGNIFICANTLY DIFFERENT FROM A WORKING COPY OF THE OTHERWISE
/ IDENTICAL CODE MOVED TO 07600-07777; FIX-UP ROUTINES THAT COME INTO PLAY
/ DURING THE BOOTSTRAP PROCESS WILL HAVE NOT BEEN APPLIED AT THE TIME OF SYSTEM
/ GENERATION; THE INTENDED CONTENTS OF BLOCK 0000 ARE NOT CAPABLE OF PERFORMING
/ PROPERLY AS A FUNCTIONING WRITE SUBROUTINE WITHOUT THESE ROUTINES HAVING BEEN
/ PERFORMED TO COMPLETE THE HANDLER IN MEMORY.

/ THE PRESENT METHOD OF SYSTEM GENERATION INCLUDES A ONE-SHOT FIX-UP ROUTINE TO
/ BE APPLIED TO THE MOVED HANDLER TO ALLOW IT TO FUNCTION FOR WRITE PURPOSES
/ ESSENTIALLY SYNTHESIZING THE EQUIVALENT OF THE FIX-UP ROUTINES MENTIONED ABOVE.
/ THE INTENDED IMAGE OF BLOCK 0000 IS THEN WRITTEN OUT BY THE CORRECTED ROUTINES
/ IN 07600-07777 ALONG WITH THE OTHER SYSTEM GENERATION PARTICULARS WRITING OUT
/ SYSTEM COMPONENTS TO THEIR DESIGNATED BLOCKS.

/ FUTURE SYSTEM GENERATION TECHNIQUES MAY DESIGNATE A NON-SYSTEM DEVICE HANDLER
/ FOR THE PURPOSE OF WRITING THE INITIAL WORKING SYSTEM IMAGE. P?S/8 NON-SYSTEM
/ DEVICE HANDLERS ARE ALWAYS SELF-SUFFICIENT BY DESIGN AND COULD BE USED TO
/ CREATE A VIABLE SYSTEM INTENDED FOR ALTERNATE HARDWARE. FOR EXAMPLE, A
/ TD8E-BASED SYSTEM COULD BE CREATED ON TC01/TC08 HARDWARE OR VICE-VERSA. A
/ TC12 LINCTAPE SYSTEM COULD BE CREATED ON DECTAPE AND ULTIMATELY TRANSFERRED
/ ONTO LINCTAPE USING PDP-12 UTILITIES [SUCH AS PRTC12F, A DECTAPE FROM/TO
/ LINCTAPE COPYING PROGRAM] TO CREATE THE FINAL WORKING SYSTEM.